UNIT – 3

ASSESSMENT – 1

SOLUTIONS

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1. **a) Propositional Logic**

Let:

- **F** = Patient A has Fever
- **C** = Patient A has Cough
- **R** = Patient A has Rash
- **B** = Patient A has Breathlessness
- **FL** = Patient A has Possible Flu
- **M** = Patient A has Possible Measles
- **P** = Patient A has Possible Pneumonia

Rules

Rule 1:

Rule 2:

Rule 3:

Facts

(b) Forward Chaining

Initial Facts

Fever = True

Cough = True

Breathlessness = True

Step 1

Rule 1:

Fever AND Cough → Flu

Both Fever and Cough are true.

Therefore:

Possible Flu = True

Step 2

Rule 2:

Fever AND Rash → Measles

Rash is not known to be true.

Therefore:

Rule 2 cannot fire.

Step 3

Rule 3:

Cough AND Breathlessness → Pneumonia

Both conditions are true.

Therefore:

Possible Pneumonia = True

Final Derived Facts

Fever

Cough

Breathlessness

Possible Flu

Possible Pneumonia

(c) Backward Chaining

Goal

Patient A has Pneumonia

Look for a rule that produces Pneumonia.

Cough AND Breathlessness → Pneumonia

Therefore, the new subgoals are:

Cough?

Breathlessness?

Both are already facts.

Therefore:

Cough = TRUE

Breathlessness = TRUE

Hence:

Pneumonia = TRUE

Goal Reduction Tree

```
          Pneumonia?
              |
     Cough AND Breathlessness
        /           \
    Cough?      Breathlessness?
      |             |
    TRUE          TRUE
```

```
      \        /
       \      /
       Pneumonia
         TRUE
```

2. **(a) Unification**

Rule 1

CNF:

Facts

Match:

Vehicle(x)

Vehicle(Ambulance)

Substitution:

Similarly:

EmergencyType(x)

EmergencyType(Ambulance)

Therefore the MGU is:

After substitution:

Rule 2

and:

Therefore:

Rule 3

Facts:

Vehicle(CarA)

Behind(CarA, Ambulance)

GreenSignal(Ambulance)

Therefore:

Resolution Proof

Clause 1

Clause 2

Clause 3

Facts

Negated Goal

Goal:

Negation:

Resolution Step 1

From Rule 1 and:

we obtain:

Resolve with:

Therefore:

Step 2

Using Rule 2:

and:

gives:

Step 3

Rule 3:

Using the three facts:

we derive:

Step 4

Resolve:

with:

Result:

Therefore:

is proved.

(c) Two Emergency Vehicles

The current KB is **not sufficient to fully handle two simultaneous emergency vehicles** because it only explicitly contains one emergency vehicle:

Ambulance

For another emergency vehicle, we need facts such as:

Vehicle(FireTruck)

EmergencyType(FireTruck)

The rules themselves are general because they use:

Therefore, the existing rules can handle multiple vehicles **if appropriate facts are added**.

Additional facts could be:

and traffic-position facts such as:

3. **(a) CNF Conversion**

P1

Eliminate implication:

So:

P2

Eliminate implication:

Apply De Morgan's law:

Therefore:

P3

Eliminate implication:

Using De Morgan's law:

Therefore:

Facts

(b) Resolution Refutation

Goal

Negate the goal:

Step 1

C1:

Fact:

Resolve:

Step 2

C2:

We have:

and:

Therefore:

Step 3

We added the negated goal:

Resolve:

with:

Therefore:

Resolution Tree

SoilDry

|

| C1

↓

IrrigationNeeded

|

| C2 + CropWheat

↓

ApplyDripMethod

|

| + ¬ApplyDripMethod

↓

□

(c) Remove SoilDry

If:

SoilDry

is removed:

We cannot derive

because P1 requires SoilDry.

Therefore we cannot use P2 to derive:

So:

ApplyDripMethod = Not Proven

Does CropAtRisk follow?

No.

P3 requires:

But merely removing the fact:

SoilDry

does **not** mean:

¬ApplyDripMethod

Therefore:

also **does not follow**.

This is an important logical distinction:

Absence of a fact does not mean its negation is true.